

14. Backup and Restore (tar & gzip)

Complete step-by-step shell commands, options explanation, and verification procedures for Linux archiving and compression tasks.

Task 1: Create a compressed backup of the Music folder

Task Requirement: Create a backup of your Music folder (or any folder with at least 3 files) using the tar command and compress it with gzip, saving the output as `music_backup.tar.gz` in your home directory.

```
# Change to home directory context and archive Music directory
tar -czvf ~/music_backup.tar.gz -C ~ Music
```

Flag & Parameter Explanation:

- **-c:** Create a new archive.
- **-z:** Compress the archive using gzip (.gz).
- **-v:** Verbose mode; displays each file being processed in real-time.
- **-f:** Specifies the archive file name (must immediately precede filename).
- **-C ~ Music:** Switches directory to user home (~) before archiving Music, avoiding full absolute paths in the archive.

Task 2: Extract archive into a new directory and verify contents

Task Requirement: Extract the contents of a tar.gz file (for example, `music_backup.tar.gz`) into a new directory called `restored_music` and verify that all files are restored correctly.

```
# 1. Create the destination directory
mkdir -p ~/restored_music

# 2. Extract into restored_music directory
tar -xzf ~/music_backup.tar.gz -C ~/restored_music

# 3. Verify extracted contents recursively
ls -lR ~/restored_music
```

Flag & Parameter Explanation:

- **-x:** Extract files from an existing archive.

- **-C ~/restored_music:** Directs tar to unpack all files into the specified target folder instead of the current working directory.
- **ls -lR:** Recursively lists files, permissions, and sizes to ensure complete restoration.

Task 3: Filtered compressed backup using find and --files-from

Task Requirement: Write a shell command that creates a compressed backup of all .jpg images from your Downloads folder into a file named images_backup.tar.gz, excluding any files larger than 5MB (Hint: Use the find command with size filtering and tar with --files-from).

```
# Search for .jpg files < 5MB and pipe list directly into tar
find ~/Downloads -type f -name "*.jpg" -size -5M | tar -czvf ~/images_backup.tar.gz -
-files-from=-
```

Flag & Parameter Explanation:

- **find ~/Downloads:** Recursively scans the Downloads folder for matching criteria.
- **-type f:** Matches regular files only (ignores sub-folders).
- **-name "*.jpg":** Matches files ending with .jpg extension (case-insensitive alternative: -iname).
- **-size -5M:** Filters for files strictly smaller than 5 Megabytes.
- **--files-from=-:** Tells tar to read the file list directly from standard input (pipe |).

Task 4: Simulate restore scenario (selective file recovery)

Task Requirement: Simulate a restore scenario: delete one file from your original Music folder, then use your music_backup.tar.gz archive to restore only the deleted file without overwriting the rest.

```
# Step 1: Simulate accidental deletion of a single file
rm ~/Music/song1.mp3

# Step 2: List contents of archive to confirm relative path of file
tar -tzvf ~/music_backup.tar.gz

# Step 3: Restore ONLY the deleted file into the original location
tar -xzvf ~/music_backup.tar.gz -C ~ Music/song1.mp3
```

Flag & Parameter Explanation:

- **tar -tzvf:** Lists ('table of contents') archive members without extracting them to inspect paths.
- **Music/song1.mp3:** Specifies the exact member path to extract. tar selectively unpacks only this file and leaves all other existing files untouched.